Mini Project
Kinematic calibration

Team members

Last name	First name	Last name	First name	Students, *third if any*

Prepared by:

Name	Signature	Date	Supervisor

Controlled by:

Name	Signature	Date	First team member

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Degree Programme
Systems Engineering

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User Requirements Specification

1. Kinematic synchronization

URS ID	Description
1.1	The machine shall synchronize all three axes during the PackML Starting state.
1.2	Synchronization only occurs in production mode.
1.3	The operator shall not have to do any action other than pressing the start button to synchronize the axes.
1.4	If the machine passes thru Stopping or Aborting, the synchronization shall be disabled.

2. Camera calibration

URS ID	Description
2.1	The machine shall not be able to perform a pick and place sequence until the camera position on the QR code has been validated by the operator.
2.1	The operator validates the calibration only once, after the start/restart of the PLC program.
2.3	The robot automatically moves to the calibration position and launches the calibration validation image upon first entering the PackML Execute state.

3. Pick and place sequences

URS ID	Description
3.1	Pick and place sequences are launched manually from the Node-RED interface.

4. Fast commissioning, with functional specifications.

URS ID	Description
4.1	You shall demonstrate your ability to recalibrate the system using only the user interface if the plate position is changed. solely by an action on PCS parameters.
4.2	The recalibration shall be made in less than five minutes.
4.3	If the PCS settings are modified, the camera calibration validation shall be reset.